



Surge S2 chipset at MWC 2018

The Surge S2 will be built on the TSMC 16 nm process and will power the Xiaomi Mi 6X, also expected to debut in February. <http://digit.in/SurgeS2>



New entry level Macbooks

Apple to release new entry-level 13-inch MacBook in the second half of 2018, likely replacing MacBook Air. <http://digit.in/NewMacS13>

About time!

Before everything,
there was time. And with
us, came the need to know
exactly what time it was

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O

nce upon a time, there was a time when no one bothered about time. Sounds nice, doesn't it? Just like a

wise wizard once said, "A wizard is never late, nor is he early, he arrives precisely when he means to." Something like that would only be possible before humankind decided that there's a need to know exactly what time of the day it was. There were several reasons behind it – as human civilisation gradually came into existence over the years, complicated activities like farming, craftsmanship, transportation and more required people to keep track of time. And as it is with most inventions, this necessity led to the development and subsequent refinement of what we know today as timekeeping. To understand this evolution, there is a certain amount of time travel into historical records that is required, that will often encounter degrees of inaccuracy not too alien to the act of timekeeping itself.

SUNNY SIDE UP

One of the earliest indicators of time that humans observed was the most significant celestial body in the sky –

the sun. While Sumerians are often credited to be the first civilisation to properly get into timekeeping using the sun, the accounts are mostly speculative as compared to the evidence-based idea that it was the Egyptians who started it all.

To broadly keep track of the passing of the day, the Egyptians around 3500 BC built tall vertical structures known as obelisks, placed at well thought out locations where the shadow could be suitably tracked. Initially, they were



Obelisks were strategically located so that the length of the shadow could indicate the time of the day

only good for keeping track of the noon, as indicated by the time when the shadow was the shortest. But eventually, it was also used to track the longest and shortest days of the year based on the longest and shortest shadows it created. Further subdivisions were also marked around it to identify smaller sections of the day. This is the idea that eventually gave rise to the sundial.

The sundial was designed to have an axis that pointed towards the celestial north pole, around which the sun appeared to rotate. As the understanding of shadow formation and light evolved, so did the sundial, which could eventually accommodate for seasonal changes in the sun's patterns. But there was still a major problem – how to keep track of time at night?

STAR LIGHT, STAR BRIGHT

Even when it came to timekeeping in the dark, the Egyptians turned out to be the brightest of the lot. Developed around 600 BC, Merkhets were more accurate than sundials and could also be used at night. A pair of merkhets aligned with the pole star would be used to determine the north, and as specific stars would cross the meridian created by them, time could be measured at night. That is, if there weren't any clouds.

WATERED DOWN

Water clocks were the first timekeeping devices that did not depend on the movement of celestial bodies for its objective (technically they still depended on the gravitational force generated by earth's rotation, but we'll take the liberty of discounting that here).

Clepsydras (Greek for 'water thief') also had Egyptian origins, with one of the very first being found in the tomb of Egyptian King Amenhotep I, who was buried around 600 BC. The name came into use when the Greeks started using them around 325 BC. The idea was to keep track of time by tracking a near constant flow of water – either by tracking the rise or fall of the level of water in a vessel.